

Bayesian Analysis

Master in Data Science for Economics

Bayesian Portfolio Optimization

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Contents

1	Introduction	2
2	Data Description	3
3	The Model	5
3.1	Likelihood	5
3.2	Prior Distributions	5
3.3	Posterior and Full Conditional Distributions	7
3.4	Gibbs Sampler	8
3.5	Optimal Portfolio Weights	8
3.6	Predictive Distribution	9
4	Empirical Results	9
4.1	MCMC Diagnostics	9
4.2	Posterior Distributions	11
4.3	Optimal Portfolio Weights	11
4.4	Predictive Distribution	12
5	Conclusion	13

1 Introduction

Constructing an optimal portfolio in the presence of uncertainty about future returns is one of the fundamental problems of modern finance.

The starting point of the literature is the framework of Markowitz (1952), which revolutionised financial analysis by introducing the concept of mean-variance optimisation (MVO): given a vector of expected returns $\boldsymbol{\mu}$ and a covariance matrix $\boldsymbol{\Sigma}$, the optimal weights maximise the expected return for a given level of risk. Before this contribution, as Kolm, Tütüncü and Fabozzi (2014) observe, the financial literature treated the relationship between risk and return in a non-systematic way, focusing on the value of individual investments rather than on their interaction within a portfolio.

The classical framework, however, presents a fundamental limitation in practice. It treats $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ as known parameters, estimated directly from historical data. Kolm et al. (2014) show that this approach makes the optimiser extremely sensitive to estimation errors, so much so that some authors call it an *error maximiser*. In particular, errors in the estimates of expected returns have a greater impact on optimal weights than errors in the covariance matrix (Chopra and Ziemba, 1993, as cited in Kolm et al., 2014). The practical result is a portfolio that is over-concentrated and unstable, with poor robustness out of sample.

To address this problem, Black and Litterman (1992) proposed a Bayesian approach that treats $\boldsymbol{\mu}$ as a random variable rather than a fixed parameter. The model uses a prior based on market equilibrium: starting from the observed weights of the market portfolio, and assuming that the market is in equilibrium (i.e. prices already summarise all available information), the Capital Asset Pricing Model is applied in reverse to recover the values of $\boldsymbol{\mu}$ that would have justified those weights. This prior is then combined with the investor's views on future returns, producing more intuitive and diversified portfolios. As the quote attributed to Markowitz himself in Kolm et al. (2014) puts it: “*the rational investor is a Bayesian*”.

In this project, this Bayesian logic is extended to the asset allocation problem across four ETFs representative of the main asset classes: US equities (SPY, SPDR S&P 500 ETF), gold (GLD, SPDR Gold Shares), long-term US government bonds (TLT, iShares 20+ Year Treasury Bond ETF), and emerging market equities (EEM, iShares MSCI Emerging Markets ETF).

Monthly returns are assumed to be distributed as a multivariate Normal with uncertain parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Independent priors are specified: Normal on $\boldsymbol{\mu}$ and Inverse-Wishart on $\boldsymbol{\Sigma}$. Because this choice of prior makes the posterior distribution unavailable in closed form (the posterior is proportional to an expression that does not belong to any known parametric family), inference is carried out via a Gibbs sampler algorithm implemented

in R.

A central contribution of the project is the specification of the informative prior on μ : the long-run expected return forecasts published by Research Affiliates (2026) are used as the centre of the prior, thereby incorporating forward-looking information not contained in the historical data of the estimation period. The results obtained under this informative prior are compared with those obtained under a diffuse prior, evaluating how sensitive the posterior estimates and optimal portfolio weights are to the choice of prior.

2 Data Description

The data used in this project consist of monthly returns on four Exchange-Traded Funds (ETFs) listed in the United States, downloaded from Yahoo Finance via the `quantmod` package in R for the period January 2010 to December 2024, for a total of 180 monthly observations.

The four ETFs were selected to represent the main asset classes:

- **SPY** (SPDR S&P 500 ETF Trust), which tracks the S&P 500 index and represents large-cap US equities;
- **GLD** (SPDR Gold Shares), which tracks the spot price of gold and is traditionally regarded as a safe-haven asset;
- **TLT** (iShares 20+ Year Treasury Bond ETF), which tracks an index of long-term US government bonds and is sensitive to changes in interest rates;
- **EEM** (iShares MSCI Emerging Markets ETF), which tracks the MSCI Emerging Markets index, providing exposure to equity markets in developing economies.

Monthly returns are computed as simple returns on the adjusted closing price at the end of each month:

$$r_{i,m} = \frac{P_{i,\text{end}(m)} - P_{i,\text{end}(m-1)}}{P_{i,\text{end}(m-1)}} \quad (1)$$

where $r_{i,m}$ is the return of asset i in month m , and $P_{i,\text{end}(m)}$ denotes the adjusted closing price of asset i on the last trading day of month m .

The dataset is split into two sub-periods with distinct roles in the Bayesian model. **Period A** (January 2010 to December 2019, $T_A = 120$ months) is used exclusively to construct the informative prior on Σ , estimating the historical risk and correlation structure among assets in a pre-pandemic period. **Period B** (January 2020 to December 2024, $T_B = 60$ months) constitutes the estimation sample on which the posterior distribution is updated via the Gibbs sampler. This temporal separation ensures that the prior and the likelihood are built on independent information.

Table 1 reports descriptive statistics of monthly returns for Period B.

Table 1: Descriptive statistics, monthly returns 2020–2024

ETF	Monthly mean	Annualised mean return	Monthly std. dev.
SPY	1.15%	13.78%	5.29%
GLD	0.96%	11.47%	4.16%
TLT	−0.62%	−7.42%	4.55%
EEM	0.03%	0.31%	5.27%

The 2020–2024 period was characterised by exceptional market conditions. US equities (SPY) benefited from a rapid post-pandemic recovery, recording an annualised mean return of 13.78%. Gold (GLD) performed well (11.47% per annum). Long-term bonds (TLT) suffered significant losses (−7.42% per annum). Emerging markets (EEM) recorded a near-zero annualised mean return of 0.31%.

Table 2: Correlation matrix, Period B (2020–2024)

	SPY	GLD	TLT	EEM
SPY	1.000	0.248	0.340	0.705
GLD	0.248	1.000	0.356	0.394
TLT	0.340	0.356	1.000	0.340
EEM	0.705	0.394	0.340	1.000

Table 3: Annualised mean returns, Period A (2010–2019)

ETF	Annualised mean return
SPY	11.27%
GLD	3.94%
TLT	4.90%
EEM	2.18%

Table 4: Correlation matrix, Period A (2010–2019)

	SPY	GLD	TLT	EEM
SPY	1.000	0.026	−0.484	0.755
GLD	0.026	1.000	0.233	0.273
TLT	−0.484	0.233	1.000	−0.358
EEM	0.755	0.273	−0.358	1.000

A comparison between the two periods reveals a significant structural break in correlations. In Period A the correlation between SPY and TLT was −0.484, confirming the

historical hedging relationship between equities and long-term bonds. In Period B this correlation reverses to +0.340, as both asset classes declined simultaneously. This instability in the correlation structure further justifies the use of Period A as the basis for the prior on Σ . Incorporating long-run structural information prevents the model from relying exclusively on an estimation sample characterised by exceptional market conditions.

3 The Model

3.1 Likelihood

Let $\mathbf{y}_t \in \mathbb{R}^4$ be the vector of monthly returns of the four ETFs at time t , with $t = 1, \dots, T$ and $T = 60$. The observations are assumed to be independent and identically distributed according to a multivariate Normal distribution:

$$\mathbf{y}_t \mid \boldsymbol{\mu}, \boldsymbol{\Sigma} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad t = 1, \dots, T \quad (2)$$

where $\boldsymbol{\mu} \in \mathbb{R}^4$ is the vector of expected monthly returns and $\boldsymbol{\Sigma} \in \mathbb{R}^{4 \times 4}$ is the covariance matrix, symmetric and positive definite by construction. The likelihood function for the full sample is:

$$f(Y \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) \propto |\boldsymbol{\Sigma}|^{-T/2} \exp\left(-\frac{1}{2} \sum_{t=1}^T (\mathbf{y}_t - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{y}_t - \boldsymbol{\mu})\right) \quad (3)$$

The i.i.d. assumption implies that the process which generated the historical returns is the same process that will generate future returns, an assumption discussed in the conclusions.

3.2 Prior Distributions

The parameters $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are uncertain and are treated as random variables. Independent priors are specified:

$$\pi(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \pi(\boldsymbol{\mu}) \cdot \pi(\boldsymbol{\Sigma}) \quad (4)$$

This choice, unlike the conjugate Normal-Inverse-Wishart prior where the prior on $\boldsymbol{\mu}$ depends on $\boldsymbol{\Sigma}$, makes the posterior distribution unavailable in closed form, so MCMC methods are needed for inference.

Prior on $\boldsymbol{\mu}$, Multivariate Normal

The natural choice for a vector of continuous parameters is the multivariate Normal, which is defined on all of $\mathbb{R}^4$ and is conjugate with respect to the Normal likelihood:

$$\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Lambda}_0) \quad (5)$$

For the **informative prior**, $\boldsymbol{\mu}_0$ is the vector of long-run expected return forecasts published by Research Affiliates (2026), converted from annual to monthly frequency using the compounding formula:

$$\mu_{0,i}^{\text{monthly}} = (1 + \mu_{0,i}^{\text{annual}})^{1/12} - 1 \quad (6)$$

These forecasts are independent of the Period B data for three reasons: they come from an external source, they have a 10-year horizon different from the estimation sample, and they are published in 2026, after the estimation period, already incorporating awareness of the exceptional conditions of 2020–2024.

The covariance matrix of the prior is $\boldsymbol{\Lambda}_0 = \sigma_\mu^2 \cdot I_4$ with $\sigma_\mu = 0.002$ (0.2% monthly). This reflects moderate confidence in the analyst forecasts: the 95% credible interval for each $\mu_{0,i}$ is $\mu_{0,i} \pm 0.4\%$ monthly, tight enough to be informative but not so tight as to overwhelm the data. Table 5 reports the prior means and the corresponding 95% credible intervals.

Table 5: Informative prior on $\boldsymbol{\mu}$: means and 95% credible intervals

ETF	Annual μ_0	Monthly μ_0	95% CI (monthly)
SPY	3.5%	0.287%	$[-0.113\%, +0.687\%]$
GLD	5.0%	0.407%	$[+0.007\%, +0.807\%]$
TLT	4.9%	0.399%	$[-0.001\%, +0.799\%]$
EEM	7.3%	0.590%	$[+0.190\%, +0.990\%]$

For the **diffuse prior**, $\boldsymbol{\mu}_0 = \mathbf{0}$ and $\boldsymbol{\Lambda}_0 = (0.10)^2 \cdot I_4$, corresponding to a standard deviation of 10% monthly: a 95% interval of $\pm 20\%$ monthly, substantially uninformative relative to the scale of the data. In this case the posterior is determined almost entirely by the likelihood.

Prior on $\boldsymbol{\Sigma}$, Inverse-Wishart

The Inverse-Wishart is the natural choice for a covariance matrix because it generates symmetric positive definite matrices by construction, and it is the conjugate prior for the covariance matrix of a multivariate Normal. This conjugacy property ensures that the full conditional of $\boldsymbol{\Sigma}$ is available in closed form:

$$\boldsymbol{\Sigma} \sim iW(\nu_0, S_0) \quad (7)$$

The mean of the Inverse-Wishart is $\mathbb{E}[\boldsymbol{\Sigma}] = S_0/(\nu_0 - p - 1)$, which is used to calibrate the parameters.

For the **informative prior**, the degrees of freedom are $\nu_0 = p + 2 + T_A = 126$, where $T_A = 120$ is the number of observations in Period A. This choice follows the same logic

as the updating formula $\nu_n = \nu_0 + T$ applied constructively: starting from the minimum $p + 2$ and adding T_A as if 120 historical observations from Period A had already been observed, giving the prior a weight equivalent to a 10-year sample.

The scale matrix is $S_0 = \hat{\Sigma}_A \cdot (\nu_0 - p - 1)$, where $\hat{\Sigma}_A$ is the sample covariance matrix of Period A (2010–2019). This ensures that the prior mean equals the pre-pandemic historical covariance:

$$\mathbb{E}[\Sigma] = \frac{S_0}{\nu_0 - p - 1} = \hat{\Sigma}_A \quad (8)$$

For the **diffuse prior**, $\nu_0 = p + 2 = 6$, the minimum required for a proper distribution, and $S_0 = 0.001 \cdot I_4$. The latter is a near-zero matrix consistent with the scale of the data (monthly variances of order 0.002) that imposes no particular covariance structure.

3.3 Posterior and Full Conditional Distributions

By Bayes' theorem:

$$\pi(\boldsymbol{\mu}, \Sigma \mid Y) \propto f(Y \mid \boldsymbol{\mu}, \Sigma) \cdot \pi(\boldsymbol{\mu}) \cdot \pi(\Sigma) \quad (9)$$

With independent priors, this distribution does not belong to any known parametric family and cannot be sampled directly. The full conditional distributions, however, are available in closed form.

Full conditional for $\boldsymbol{\mu} \mid \Sigma, Y$

The product of the Normal likelihood and the Normal prior on $\boldsymbol{\mu}$ is again Normal, by the conjugacy of the Normal family with respect to itself:

$$\boldsymbol{\mu} \mid \Sigma, Y \sim \mathcal{N}(\boldsymbol{\mu}_n, \Lambda_n) \quad (10)$$

with updated parameters:

$$\Lambda_n = \left(\Lambda_0^{-1} + T \cdot \Sigma^{-1} \right)^{-1} \quad (11)$$

$$\boldsymbol{\mu}_n = \Lambda_n \left(\Lambda_0^{-1} \boldsymbol{\mu}_0 + T \cdot \Sigma^{-1} \bar{\mathbf{y}} \right) \quad (12)$$

where $\bar{\mathbf{y}} = \frac{1}{T} \sum_{t=1}^T \mathbf{y}_t$ is the sample mean. The posterior mean $\boldsymbol{\mu}_n$ is a weighted average between the prior $\boldsymbol{\mu}_0$ and the data $\bar{\mathbf{y}}$, with weights determined by the prior precision Λ_0^{-1} and the data precision $T \cdot \Sigma^{-1}$ respectively.

Full conditional for $\Sigma \mid \mu, Y$

The product of the Normal likelihood and the Inverse-Wishart prior on Σ is again an Inverse-Wishart, by conjugacy:

$$\Sigma \mid \mu, Y \sim iW(\nu_n, S_n) \quad (13)$$

with updated parameters:

$$\nu_n = \nu_0 + T \quad (14)$$

$$S_n = S_0 + \sum_{t=1}^T (\mathbf{y}_t - \mu)(\mathbf{y}_t - \mu)' \quad (15)$$

The degrees of freedom increase by $T = 60$ observations from Period B, while the scale matrix is updated by adding the matrix of squared deviations of the data from the current value of μ .

3.4 Gibbs Sampler

Since both full conditionals can be sampled directly, inference is conducted via a Gibbs sampler algorithm. Starting from an initial value $\Sigma^{(0)} = S_0/(\nu_0 - p - 1) = \hat{\Sigma}_A$, the algorithm alternates the following steps for $s = 1, \dots, S$:

1. Compute Λ_n and μ_n using $\Sigma^{(s-1)}$; sample $\mu^{(s)} \sim \mathcal{N}(\mu_n, \Lambda_n)$
2. Compute S_n using $\mu^{(s)}$; sample $\Sigma^{(s)} \sim iW(\nu_n, S_n)$

Since the Inverse-Wishart distribution is not directly available in base R, sampling is performed by inverting a draw from the Wishart distribution:

$$W \sim \mathcal{W}(\nu_n, S_n^{-1}) \implies \Sigma = W^{-1} \sim iW(\nu_n, S_n) \quad (16)$$

The algorithm is run for $S = 10,000$ iterations. The first $S_{\text{burn}} = 2,000$ samples are discarded as burn-in: the initial period in which the chain has not yet reached its stationary distribution, where the samples reflect the starting point more than the posterior. The remaining $S^* = 8,000$ samples constitute the posterior sample.

3.5 Optimal Portfolio Weights

The optimal portfolio weights are computed by solving the mean-variance optimisation problem for each of the $S^* = 8,000$ posterior samples:

$$w^{(s)} = \arg \max_w \left\{ \mathbf{w}' \mu^{(s)} - \frac{\gamma}{2} \mathbf{w}' \Sigma^{(s)} \mathbf{w} \right\} \quad \text{s.t.} \quad \sum_i w_i = 1, \quad w_i \geq 0 \quad (17)$$

with risk aversion coefficient $\gamma = 3$, the standard value in the literature for an investor with moderate risk aversion. The optimal Bayesian portfolio is the average of the weights across samples:

$$\bar{\mathbf{w}} = \frac{1}{S^*} \sum_s \mathbf{w}^{(s)} \quad (18)$$

This approach, known as the distribution of optimal weights (Avramov and Zhou, 2010), propagates parameter uncertainty directly into the portfolio weights, producing more conservative and diversified portfolios than classical Markowitz optimisation.

3.6 Predictive Distribution

The predictive distribution for the future return $\mathbf{y}_{T+1}$ integrates out uncertainty over the model parameters:

$$f(\mathbf{y}_{T+1} | Y) = \int f(\mathbf{y}_{T+1} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) \cdot \pi(\boldsymbol{\mu}, \boldsymbol{\Sigma} | Y) d\boldsymbol{\mu} d\boldsymbol{\Sigma} \quad (19)$$

This distribution is possible to construct thanks to the i.i.d. assumption: the parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ estimated on Period B describe the same process that will generate future returns. The integral is approximated by simulation. For each posterior sample $(\boldsymbol{\mu}^{(s)}, \boldsymbol{\Sigma}^{(s)})$ a future return is generated:

$$\mathbf{y}_{T+1}^{(s)} \sim \mathcal{N}(\boldsymbol{\mu}^{(s)}, \boldsymbol{\Sigma}^{(s)}) \quad (20)$$

The 8,000 simulated values $\{\mathbf{y}_{T+1}^{(s)}\}_{s=1}^{8000}$ constitute the approximate predictive distribution, from which the median (50th percentile) and 95% predictive credible intervals (2.5th and 97.5th percentiles) are computed via empirical quantiles.

4 Empirical Results

4.1 MCMC Diagnostics

Before analysing the empirical results, the convergence of the Gibbs sampler is verified using three diagnostic tools.

Trace plots

The trace plots of the posterior samples of $\boldsymbol{\mu}$ show well-mixing chains for all four ETFs, with irregular oscillations around a stable value, no upward or downward trends, and no sections in which the chain becomes stuck. The plots indicate a well mixing chain with similar means and variances in different sections of the chain, evidence of good convergence and successful completion of the transient (burn-in) stage.

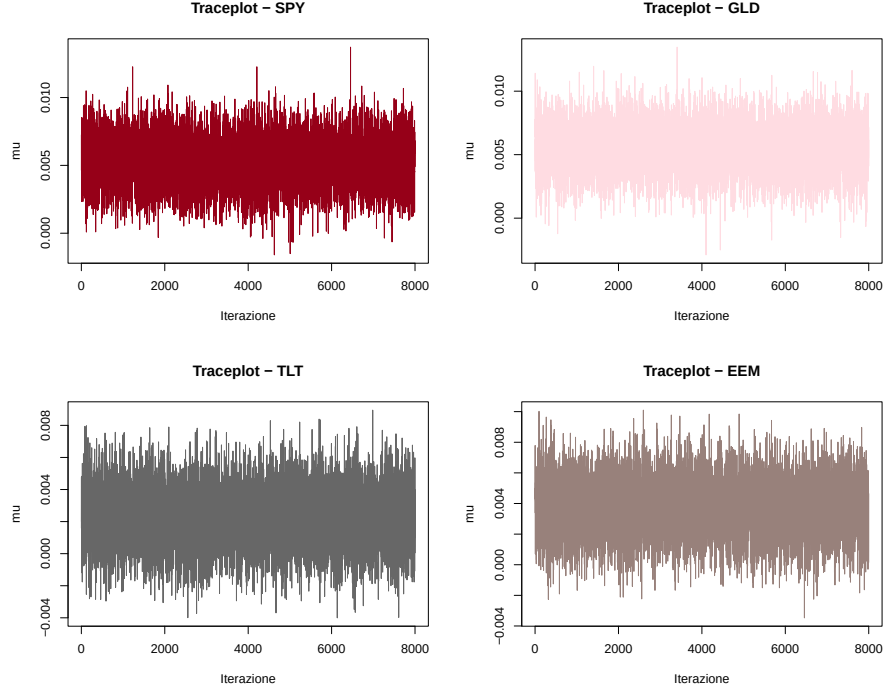


Figure 1: Trace plots of posterior samples of μ , informative prior.

Effective random sample size (N_{eff})

The effective random sample size N_{eff} measures the equivalent number of independent samples, accounting for the autocorrelation present in the MCMC output. Values of N_{eff} close to $N = 8,000$ indicate negligible autocorrelation, so that no thinning is required. Table 6 reports the values obtained.

Table 6: Effective random sample size (out of 8,000 posterior draws)

ETF	N_{eff} , Informative prior	N_{eff} , Diffuse prior
SPY	8000	7378
GLD	8000	8238
TLT	8000	7748
EEM	8000	8000

N_{eff} is close to N for both priors, indicating negligible autocorrelation among the samples and confirming that no thinning is needed. The informative prior produces $N_{\text{eff}} = 8,000$ for all four parameters.

Gelman-Rubin Diagnostic

The Gelman-Rubin diagnostic is conducted by running two independent chains ($m = 2$) from overdispersed starting values (`set.seed(123)` and `set.seed(456)`). The potential

scale reduction factor $\hat{R}$ is equal to 1.00 for all four parameters, with a 97.5% confidence interval of 1.00. Values of $\hat{R}$ below 1.1 indicate convergence: the results confirm that both chains have converged to the same posterior distribution.

4.2 Posterior Distributions

Table 7 reports the posterior means of annualised expected returns and the corresponding 95% credible intervals for the two prior specifications, compared with the analyst forecasts and the historical data from Period B.

Table 7: Posterior means and 95% credible intervals (annualised returns)

ETF	Analyst	Data 2020–24	Post. inf.	CI 95% inf.	Post. dif.	CI 95% dif.
SPY	3.5%	13.78%	6.34%	[2.09%, 10.57%]	13.71%	[−2.07%, 30.21%]
GLD	5.0%	11.47%	6.60%	[2.12%, 11.01%]	11.49%	[−1.01%, 23.95%]
TLT	4.9%	−7.42%	2.74%	[−1.67%, 7.05%]	−7.41%	[−21.16%, 6.54%]
EEM	7.3%	0.31%	4.45%	[0.12%, 8.77%]	0.34%	[−15.16%, 16.12%]

The posterior mean μ_n is a weighted average between the analyst forecast μ_0 and the sample mean $\bar{y}$, with weights determined by the precision of the prior and the precision of the data. The results show a clear regularisation effect of the informative prior.

For SPY, the posterior mean of 6.34% represents a compromise between the analyst forecast (3.5%) and the historical return of Period B (13.78%). The effect is even more pronounced for TLT: the informative prior brings the posterior mean to 2.74%, moving it substantially away from the historical figure of −7.42%. The prior recognises that the losses on long-term bonds in Period B reflect an exceptional context rather than a structural return.

The diffuse prior instead produces posterior means almost identical to the historical data of Period B, with much wider credible intervals, a direct consequence of the lower information incorporated. For example, the 95% interval for TLT under the diffuse prior is [−21.16%, 6.54%], much wider than the interval [−1.67%, 7.05%] under the informative prior. This comparison illustrates how the choice of prior has a substantial impact on inference when the estimation sample is limited and characterised by exceptional market conditions.

4.3 Optimal Portfolio Weights

Table 8 reports the optimal portfolio weights for the three specifications, computed via quadratic programming using the `quadprog` package in R.

Table 8: Optimal portfolio weights

ETF	Markowitz	Post. diffuse	Post. informative
SPY	53.1%	50.5%	43.9%
GLD	46.9%	49.1%	41.2%
TLT	0.0%	0.1%	10.3%
EEM	0.0%	0.3%	4.6%

The Bayesian portfolio with informative prior is the most diversified, assigning positive weights to all four assets. Compared with the diffuse prior, which produces an allocation almost identical to classical Markowitz, the informative prior reduces the concentration on SPY and GLD and increases the weight of TLT to 10.3% and EEM to 4.6%, reflecting the Bayesian correction towards more balanced long-run expected returns.

It is worth noting that EEM receives the lowest weight despite having the highest expected return according to Research Affiliates (7.3%). This is consistent with the logic of the model: EEM exhibits the highest volatility in the sample (monthly standard deviation of 5.27%) and a correlation of 0.705 with SPY, which makes over-allocation to this asset costly in risk terms. TLT, despite a lower expected return, contributes more to diversification precisely because it is less correlated with the other assets.

4.4 Predictive Distribution

Table 9 reports the median and 95% predictive credible intervals for annualised future returns, obtained by simulating $\mathbf{y}_{T+1}^{(s)} \sim \mathcal{N}(\boldsymbol{\mu}^{(s)}, \boldsymbol{\Sigma}^{(s)})$ for each of the 8,000 posterior samples.

Table 9: 95% predictive credible intervals (annualised returns)

ETF	Median inf.	CI 95% inf.	Median dif.	CI 95% dif.
SPY	6.02%	[−96.8%, +108.7%]	13.93%	[−113.6%, +140.2%]
GLD	6.79%	[−98.8%, +113.9%]	11.45%	[−87.1%, +111.1%]
TLT	3.01%	[−88.1%, +99.4%]	−8.38%	[−116.1%, +101.4%]
EEM	4.13%	[−120.5%, +129.6%]	0.88%	[−125.5%, +125.7%]

The predictive intervals are wide, as expected for forecasts of monthly returns on highly volatile financial assets. The predictive distribution incorporates two sources of uncertainty: parameter uncertainty, captured by the posterior, and the intrinsic variability of future returns given the model structure. The informative prior produces more moderate medians and narrower intervals than the diffuse prior, in particular for TLT (3.01% vs −8.38%), confirming the impact of the prior on future predictions as well.

5 Conclusion

The Bayesian inference framework has been applied to the asset allocation problem across a portfolio of four ETFs representative of the main asset classes: US equities (SPY), gold (GLD), long-term government bonds (TLT) and emerging market equities (EEM). The model assumes that monthly returns are distributed as a multivariate Normal with uncertain parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, specifying independent priors: Normal on $\boldsymbol{\mu}$ and Inverse-Wishart on $\boldsymbol{\Sigma}$, and conducting inference via a Gibbs sampler implemented in R.

The central contribution is the specification of an informative prior on $\boldsymbol{\mu}$ based on the long-run expected return forecasts of Research Affiliates (2026), compared with a diffuse prior.

The empirical results highlight three main findings.

First, the MCMC diagnostics confirm the quality of inference: the trace plots show well-mixing chains, the effective random sample size is close to 8,000 for all parameters, and the Gelman-Rubin diagnostic produces $\hat{R} = 1.00$, indicating optimal convergence.

Second, the informative prior has a significant regularisation effect on the posterior means. In particular, the posterior mean of TLT under the informative prior is 2.74% per annum compared to the historical figure of -7.42% . The prior recognises that the losses on long-term bonds in Period B reflect an exceptional monetary policy context rather than a structural return.

Third, the choice of prior has a direct impact on the optimal portfolio weights. The Bayesian portfolio with informative prior is more diversified, assigning positive weights to all four assets (SPY 43.9%, GLD 41.2%, TLT 10.3%, EEM 4.6%), while classical Markowitz and the diffuse prior concentrate the allocation on SPY and GLD, assigning zero weight to TLT and EEM.

A limitation of the present work is the i.i.d. assumption, which ignores the temporal dependence and time-varying volatility characteristic of financial markets. A natural extension would be to evaluate the out-of-sample forecasting performance of the model by comparing the predictive distributions with the returns realised in 2025, in order to assess whether the informative prior produces more accurate predictions than the diffuse prior.

References

- [1] Avramov, D., & Zhou, G. (2010). *Bayesian Portfolio Analysis*. Annual Review of Financial Economics, 2, 25–47.

- [2] Black, F., & Litterman, R. (1992). *Global Portfolio Optimization*. Financial Analysts Journal, 48(5), 28–43.
- [3] Chopra, V.K., & Ziemba, W.T. (1993). *The Effect of Errors in Means, Variances, and Covariances on Optimal Portfolio Choice*. Journal of Portfolio Management, 19(2), 6–11.
- [4] Kolm, P.N., Tütüncü, R., & Fabozzi, F.J. (2014). *60 Years of Portfolio Optimization: Practical Challenges and Current Trends*. European Journal of Operational Research, 234(2), 356–371.
- [5] Markowitz, H. (1952). *Portfolio Selection*. Journal of Finance, 7(1), 77–91.
- [6] Research Affiliates (2026). *Asset Allocation Interactive. Expected 10-Year Nominal Returns, as of 03/31/2026*. Retrieved from <https://interactive.researchaffiliates.com/asset-allocation>
- [7] Yahoo Finance (2024). *Historical price data for SPY, GLD, TLT, EEM*. Retrieved via quantmod R package from <https://finance.yahoo.com>